

NIKUNJ GARG

Java Backend Developer | BCA Student

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SUMMARY

BCA student specializing in Java backend development with Spring Boot 3, PostgreSQL, and Spring Security. Experienced in architecting secure RESTful APIs, real-time WebSocket communication, relational database schemas, and deploying cloud backends across competitive hackathons and production-style projects.

TECHNICAL SKILLS

- **Languages & Database:** Java 21, SQL, PostgreSQL
- **Backend Frameworks:** Spring Boot 3, Spring Security, Spring Data JPA, Hibernate, RESTful APIs, WebSockets (STOMP)
- **Security & API Standards:** JWT (Access & Refresh Token Rotation), BCrypt, Bean Validation, OpenAPI/Swagger 3.0
- **DevOps & Tools:** Docker, Docker Compose, Git, GitHub, Maven, Postman, Thunder Client, Cloudinary, Railway
- **Core Engineering:** OOP, Data Structures & Algorithms, Relational Schema Design, Clean Architecture, Unit Testing (JUnit 5)

PROJECTS

AgriSathi AI | Java 21, Spring Boot 3, PostgreSQL, Cloudinary, Swagger [\[GitHub Repo\]](#) • [\[Live Demo\]](#)

- Architected a modular Java/Spring Boot backend for an AI-powered agricultural advisory platform during a 30-day hackathon.
- Implemented secure **JWT authentication** with **Role-Based Access Control (RBAC)** supporting FARMER and BUYER roles.
- Built REST endpoints for Farmer Profiles, Crop Lifecycle Tracking, and AI Disease Diagnosis (`POST /api/v1/disease/scan`) with Cloudinary media storage and computer vision integration.
- Integrated hyper-local 7-day predictive weather forecasting APIs and documented all endpoints with interactive **OpenAPI/Swagger 3.0**.

DevTinder | Java, Spring Boot, PostgreSQL, WebSockets, Railway [\[GitHub Repo\]](#) • [\[Live Demo\]](#)

- Engineered and deployed the backend for a developer teammate matchmaking platform in 72 hours during Devlynix Buildathon 2.0.
- Designed a PostgreSQL-backed relational API layer for developer profiles, tech stack tags, and mutual connection matching requests.
- Implemented real-time bidirectional group & direct messaging using **WebSockets (STOMP protocol)** with persistent chat history.
- Configured production backend deployment on **Railway** with containerized environment variables and health check monitors.

LibroSphere | Java 21, Spring Boot 3, PostgreSQL, Spring Security, JUnit 5 [\[GitHub Repo\]](#)

- Eliminated double-checkout TOCTOU race conditions on multi-copy inventory using pessimistic database row locking (`SELECT ... FOR UPDATE`) and atomic decrements under high concurrency.
- Engineered an automated FIFO waitlist queue intercepting returned assets to reserve 48-hour pickup windows for next-in-line patrons before releasing copies to public availability.
- Architected stateless JWT authentication with database-backed **Refresh Token Rotation** (`/auth/refresh`), role-based authorization, and centralized RFC-compliant error handling.
- Built a **48-test automated test suite** (100% pass) including multi-threaded concurrency suites (`CountDownLatch` , `ExecutorService`) proving zero inventory underflow under contention.

EDUCATION

Bachelor of Computer Applications (BCA) — Shri Guru Ram Rai University, Dehradun 2025 – 2028

Cumulative CGPA: 7.48 / 10.0 • **Semester 2:** 7.96 / 10.0 (Upward Trajectory)

ACHIEVEMENTS & LEADERSHIP

- **Secretary — TechMantra Coding Club:** Promoted from Member (10 mos) to Secretary; coordinate student workshops and supported internal rounds for Smart India Hackathon (SIH).
- **Startup Innovation Recognition:** Awarded Certificate of Recognition for presenting an innovative startup idea at the Innovation & Incubation Centre (IIC), SGRR University & UCOST.
- **Java Programming Certification:** Completed Java Programming certification through the IIT Bombay Spoken Tutorial Project.